

Linking calculated and perceived accessibility to suggest equity standards

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Abstract

This is the abstract.

It consists of two paragraphs.

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1. Introduction

Transportation equity refers to the distribution of the burdens and benefits of the transportation system among the population (Karner et al., 2020, 2024; Pereira et al., 2017; Pereira & Karner, 2021). Burdens include health-related concerns arising from exposure to air pollution, noise, and traffic; sedentary lifestyles; and the risk of injury or death from collisions. In contrast, the primary benefit is the improvement of population accessibility (Pereira & Karner, 2021), understood as the ease of reaching (spatially distributed) opportunities and destinations (5–8), such as employment, healthcare, educational services, and other places for social interaction.

Accessibility is one of the primary outcomes of spatial development (Páez et al., 2012). It is influenced by the spatial distribution of the population and opportunities and depends on the characteristics and performance of the transportation network, as a product of both transportation systems and land use patterns (Wee & Geurs, 2011). There is growing consensus among researchers and transportation planners that the goal of transport policy should be to improve accessibility. This contrasts with traditional planning approaches that have prioritized mobility patterns that stimulate private automobile use and higher speeds as solutions to congestion and long commuting times. Mobility-centered policies often focus on quantitative metrics, such as increasing the number of trips, raising average speeds, and reducing congestion, treating mobility as an end in itself. However, this approach is problematic because people rarely travel simply for the sake of moving; instead, they travel to access opportunities, activities, or other people at their destinations. Therefore, since the main goal is to enable people to access activities, land use and transport planning practices must be redesigned to prioritize accessibility.

Insufficient or inadequate accessibility can prevent individuals from participating in location-based activities they wish to engage in for reasons beyond their control, in a process described in the literature as Transport-Related Social Exclusion (TRSE) (Casas, 2007; Lucas, 2012; Paez et al., 2009; Preston & Rajé, 2007; Luz & Portugal, 2022).

Insufficient accessibility, when combined with other forms of social and economic disadvantage, can result in transportation poverty (Casas, 2007; Litman, 2024; Lucas, 2012, 2019; Paez et al., 2009; Preston & Rajé, 2007): the failure of transportation and land use systems to provide access to essential services or

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employment (Luz & Portugal, 2022). Transportation poverty increases vulnerability among individuals who already face challenges associated with age (van Dülmen et al., 2022), income (Allen & Farber, 2019), race (Awaworyi Churchill, 2020), gender (Murphy et al., 2023), disability (Upham et al., 2022), among other factors (Kim et al., 2024).

With the gentrification of Canadian downtowns and the suburbanization of low-income populations, it is estimated that over one million Canadians live in transport poverty (Allen & Farber, 2019). An equity standard, when operationalized, defines the level of fairness to be adopted and the acceptable inequality (Soukhov, Anastasia et al., 2024). However, the lack of standards for the equitable distribution of transportation benefits complicates efforts to address transportation poverty.

Additionally, there is an absence of transportation data in the form of open data products to support urban planners (Arribas-Bel et al., 2021; Behrendt & Sheller, 2021; Páez, 2021), and transportation planners receive limited formal guidance in assessing the equity implications of their projects (Miller, 2018; Palm et al., 2023; te Brömmelstroet et al., 2016). Lastly, few studies explicitly link accessibility levels with employment outcomes, such as employment opportunities declined or accepted due to transport conditions (Soukhov, Anastasia et al., 2024), or individuals’ perceptions - for instance, their opinion on how easy is to reach employment opportunities and whether jobs are available within an ideal commuting time. Combined, this results in a gap in the development and implementation of accessibility standards to ensure equity in transportation planning and policy.

This study is guided by the following research questions: How can calculated job accessibility be related to perceived accessibility and employment outcomes? And could a given level of accessibility serve as a standard or goal for equitable transport planning? To answer these questions, this study has the general objective of identifying key thresholds in the relationship between calculated job accessibility and people’s perceptions of employment outcomes in order to define equity standards for transportation planning.

Our research advances the current understanding of the literature on the equitable distribution of transportation benefits by proposing a method that identifies calculated accessibility thresholds based on their relationship with perceived accessibility. Identifying minimum job accessibility levels in practice would facilitate the implementation of normative criteria that can guarantee standards to prevent equity-deserving groups from being excluded from the labour market. Furthermore, accessibility thresholds can provide policymakers with an understanding of how much accessibility is needed to meet the fundamental needs of the population, increasing the potential for government adoption and enhancing the comprehensibility and relevance of standards for community members and those experiencing unequal mobility.

We also present, in a transparent manner, the moral guidelines that support the distribution of accessibility. In addition, to facilitate collaboration and further analysis and to ensure the transparency and replicability of our study, we developed this paper using literate programming, with the data analysis code accessible through our GitHub page (<https://github.com/dias-bruno/NSTP>), in alignment with best practices in spatial data science (Arribas-Bel et al., 2021).

2. Theoretical Background

2.1. Calculated and perceived accessibility

The most commonly used accessibility measures are based on the gravity model of Hansen and make use of spatial data. This model considers the number of opportunities that a person can reach to be proportional to the sum of opportunities in each destination, discounted by the travel cost to reach that destination. This type of metric is classified as a location-based measure, representing a characteristic of a given location that assumes that all people in this location have equal access to the spatially distributed opportunities. Such measures are sensitive to land use factors and to the performance of the transportation system (accessibility 2004). However, they do not take into account the individual characteristics of people.

The perceived accessibility, on the other hand, refers to the perception of people of how easily they can reach opportunities (Lattman et al., 2016; Pot et al., 2021), resulting from not only the interaction of land-use and transport infrastructure, but also from travel options (e.g., having a car or a transit pass), travel experiences, and individual characteristics and preferences (e.g., Lattman et al., 2016; Mehdizadeh

2025). Often, perceived accessibility deviates from place-based calculated accessibility, with some research indicating that calculated accessibility might not be able to capture the multiple ways in which people experience a lack of access to opportunities (Zhu & Lucas, 2025; Paromita, 2026). Explanations for these differences are partially attributed to differences in individuals' travel attitudes, capabilities, and preferences (J. De Vos et al. 2025 - <https://www-sciencedirect-com.libaccess.lib.mcmaster.ca/science/article/pii/S0965856425001818?via%3Dihub>; van der Vlugt et al., 2022).

Even though, as demonstrated in the following subsection, calculated accessibility is still commonly used to evaluate the effect that the interaction between land-use characteristics and the transportation system has on employment outcomes. In addition, existing studies on perceived accessibility seem to have difficulties in creating standardized definitions and measurement tools, contributing to the somewhat limited knowledge on the topic (e.g., Negm et al., 2025). In this paper, as suggested by Pot et al. (2024), we define sufficient levels of accessibility based on the relationship between calculated accessibility and people's perception of employment outcomes, while also controlling for individual characteristics.

2.2. *Job accessibility and employment outcomes*

Studies connecting employment outcomes to job accessibility date back to the 1960s, when John Kain (1968) proposed the spatial mismatch hypothesis while analyzing Detroit and Chicago, USA. The author hypothesized that the low employment rates among Black Americans in inner cities resulted from employment decentralization and the difficulty faced by the Black population in moving out of these areas, primarily due to racial discrimination and segregation.

Since the work of Kain, studies linking calculated job accessibility and different employment outcomes have presented mixed results. Bania et al. (2008) found no significant effect for job access when examining its relationship with employment, earnings, hourly wage, and weekly hours worked for women leaving welfare in Ohio, USA. Delmelle et al. (2021) also found that changes in car accessibility had no significant impact on employment rates for different population income groups in the Metropolitan Area of Charlotte, North Carolina, USA, arguing that the spatial separation between lower-wage workers and their workplaces might already be relatively small. Studying Los Angeles, USA, Merlin and Hu (2017) found mixed-effects of job accessibility on employment status, with a stronger association for more highly educated groups. The authors argued that lower-educated groups may be more limited by non-spatial factors that impede their employment opportunities, such as qualifications and social skills. On the other hand, for workers with a higher level of education, the time and monetary costs of spatial separation itself may play a relatively more important role, so accessibility measures may have a greater impact on their decisions and ability to enter the labor market.

In contrast, other studies found significant effects of job accessibility on employment outcomes (Bastiaanssen et al., 2021; Duarte et al., 2023; Jin & Paulsen, 2018; Johnson et al., 2017; Matas et al., 2010). Jin and Paulsen (2017) used a difference-in-differences model for the Chicago Metropolitan Area, finding that increasing job accessibility between 2010 and 2017 was associated with reduced unemployment rates. Duarte et al. (2023) found that job accessibility had a positive effect on participation in the labour market for women, a negative effect on informal employment for women, and a positive effect on the probability of being employed for both women and men. More recently, Bastiaanssen et al. (2025) examined differential employment effects of job accessibility by public transport in combination with the bicycle for different educational groups at the national level in the Netherlands. They found that jobs for lower-educated groups are often located outside areas of high accessibility, thereby reducing job accessibility among these groups, while jobs for higher-educated groups tend to be concentrated in and around downtowns and are associated with greater accessibility.

2.3. *Equity Standards*

Accessibility standards are often operationalized as a type of opportunity-based standard, defined using thresholds in access to opportunities (e.g., the number of opportunities within a 10-minute walking trip) or based on travel impedance thresholds (e.g., a given distance to a subway station). After a systematic review, Souhkov et al. identified that this type of standard is the most commonly applied in transportation literature (68%).

Yenisetty & Bahadure (2020) set an opportunity-based standard, considering a distance of more than 1,200 meters between amenities, services, and facilities by public transport to be unfair when analyzing cities in India. Tiznado-Aitken (2018) defined unfair cases as those where bus stops and metro stations are located more than 1 kilometer away. Sun & Thakuriah adopted 400 meters as the maximum acceptable walking distance to bus and tram stops, and 800 meters to ferry stations in England. In another study, Peunghumsai et al. (2020) identified opportunity standards through the ratio of public transport supply to demand, considering conditions fair when demand is greater than supply.

However, one must to consider the possible implications of relying on normative values to define thresholds. For example, when adopting a fixed distance to a certain bus stops or train stations the justification for this choice is often unclear. In addition, the use of distance-based thresholds may be problematic, since the time required to cross a certain distance can vary considerably within a city, even for the same mode of transportation.

3. Materials and Methods

The methodology for obtaining standards of accessibility based on the relationship between calculated measures and people’s perceptions can be synthesized in six main steps: modelling job accessibility, selecting population groups with the lowest perceived accessibility, predicting employment outcomes using calculated accessibility, setting equity standards for transportation, and defining priority geographical areas for policy interventions. We recognize accessibility as the transport benefit to be distributed across society, but we also find it necessary to define the moral principles guiding such distribution, how the distribution should occur to ensure fairness, and the requirements for considering a population group a priority group. Therefore, the following guidelines have been defined:

1. We recognize accessibility as a human capability, and a higher level of accessibility implies more choice.
2. A fair transportation and land-use system provides sufficient accessibility to all members of society. A minimum level of job accessibility is essential for individuals to meet their basic needs and, while not sufficient on its own, it is a prerequisite for promoting equality of job opportunities (sufficientarian perspective).
3. The minimum level of accessibility should be defined in terms of the most unsatisfied population group in relation to the transportation system, while considering equity perspectives and implications.
4. A fair distribution of accessibility should focus on reducing inequalities of opportunity by prioritizing disadvantaged populations, while simultaneously aiming to improve overall accessibility levels.
5. Although accessibility standards should be based on the needs of disadvantaged populations, it is also necessary to identify areas where public policy interventions should be targeted, as the needs of populations can vary significantly depending on individual constraints.

3.1. Study Area and Data Sources

Our study area refers to the urban conurbation formed by the Greater Toronto Area and the Hamilton Area (GTHA), in Ontario, Canada. More specifically, the study area refers to the Census Divisions of the regional municipalities of Durham, Hamilton, Halton, Peel, Toronto, and York. According to the latest census (2021), over 7.2 million residents live in this area, which is spread across around 7,300 square kilometers.

Regarding the primary data sources, this study uses data from the 2021 Census of Population (CoP), from Statistics Canada, and from the National Survey of Transport Poverty (NSTP), from the Mobilizing Justice Partnership. The Census was selected due to its various socio-economic and demographic variables and due to the “Commuting to Work” questionnaire. The Census allows the identification of individual in the labour force and include variables required to calculate job accessibility, such as place of work, commuting modes, and commuting duration. The Census also provides standardized spatial datasets with statistical and administrative geographical boundaries, including the Census Tract (CTs), which are used in this study as the spatial units used for calculating accessibility measures. According to the Census, 3,868,120 individuals in the labour force reside in the study area, accessing a total of 3,043,730 job opportunities. Figure 2 shows the total labour force and the number of jobs across census tracts in the study area.

The second survey, NSTP, is a large-scale survey designed to understand transport-related social exclusion (TRSE) and transport poverty in Canada, collecting responses from more than 27,000 people across the country and providing an unprecedented look at the transportation challenges Canadians face, especially among equity-deserving groups whose experiences are often overlooked. The survey is openly available to researchers, decision-makers, and the public, and offers information on household dynamics, respondents' perceptions, and individual-level socioeconomic data. The NSTP provides respondent weights to enable the creation of statistics that are consistent and representative of the Canadian population (Morency et al., 2024). This survey was selected because it contains respondents' perceptions of accessibility and employment outcomes, as well as socioeconomic and demographic variables at the individual level, with a geographic resolution that enables the linking with calculated accessibility.

Not all questions from the NSTP were answered by all survey respondents. The questions were defined according to demographic and socioeconomic profiles and to previous responses. For this study, the NSTP sample corresponds to respondents residing in the GTHA who also answered the questions displayed in Table 1, totalling 2,371 respondents that represent 3,684,512 individuals in the population (weighted sample). Using a continuum scale from 0 (completely disagree) to 100 (completely agree), the respondents answered to what extent they agreed with the statements in Table 1.

Table 1: Weighted summary statistics of the outcome variables.

To what extent do you agree with the following statements?	N = 3,684,512
There are too few jobs available for me in my ideal commuting time	58.84 (27.71)
With the transport options available to me, I can easily reach work	72.64 (22.58)
I have had to decline employment opportunities because of my transport situation	41.06 (31.30)
It takes too long to travel to work	54.04 (29.26)

¹ Mean (SD)

3.2. Modelling accessibility

In this study, we implement the multimodal spatial availability measure to model job accessibility. This measure is an extension of the spatial availability measure to include travel by different modes. Both measures consider the competition for opportunities, which avoids a single job opportunity being equally available to more than one person. Competitive measures matter when considering employment outcomes because jobs are exclusive, and their use by one person prevents their use by another. In addition, these measures relies on a proportional allocation that balances the cost of travel and the population size, imposing a constrained calculation that allocates the opportunities to each origin ensuring that, when all jobs allocated to an origin are summed, they equal the total number of opportunities in the region. Lastly, both measures have the advantage of resulting in an interpretable indicator, in which the output of the model refers to the number of jobs reachable in each spatial unit.

We calculated the multimodal spatial availability V_i^m using Equation (1):

$$V_i^m = \sum_{j=1}^J O_j F_{ij}^{tm} \quad (1)$$

where:

- V_i^m is the spatial availability of opportunities at origin i and travel mode m ;
- O_j is the number of opportunities at destination j ;
- F_{ij}^{tm} is a balancing factor allocating opportunities at j to origin i considering mode m .

F_{ij}^{tm} incorporates both population demand and travel impedance, defined according to Equation 2:

$$F_{ij}^{tm} = \frac{F_i^{pm} \cdot F_{ij}^{cm}}{\sum_{m=1}^M \sum_{i=1}^N F_i^{pm} \cdot F_{ij}^{cm}} \quad (2)$$

The population-based allocation factor is obtained by:

- $F_i^{pm} = \frac{P_i^m}{\sum_m \sum_i P_i^m}$

where P_i^m represents the population at origin i using mode m .
On its turn, the travel impedance allocation factor is given by:

- $F_{ij}^{cm} = \frac{f^m(c_{ij}^m)}{\sum_m \sum_i f^m(c_{ij}^m)}$

where:

- c_{ij}^m is the cost to travel between i and j for mode m ;
- $f^m c_{ij}^m$ is an impedance function representing the effect of travel cost.

Using microdata from the 21CoP, we calculated for each CT the number of people in the labour force that commute to work by walking, cycling, car-motorized modes (those who commute by car, truck, or van as a driver or passenger, and motorcycles or scooters), and public transit (those who commute by bus, subway, light rail, streetcar, commuter train, and passenger ferry). We distributed remote workers according to the share of each transportation mode in each CT. Then, we summed the number of jobs in each CT considering the place of work mentioned by each worker.

We calculated the travel times between origins and destinations for each travel mode using the R package `r5r` with a street network obtained from OpenStreetMap. We used the General Transit Feed Specification (GTFS) obtained from the Canadian Public Transit Network Database to retrieve the public transit schedule and coverage. Considering that our study area contains different transit agencies, we evaluated the GTFS files to identify a common date/week with public transit service. After this analysis, we set 8 a.m. (morning peak time) of December 22, 2025, as the trip departure time. We also used Digital Elevation Models (DEM) from the Canadian Digital Surface Model (CDSM), with a spatial resolution of about 20 meters, to estimate the effect of declivity on walking and cycling times. We set a maximum duration of 180 minutes (3 hours) for car and public transit trips, with a maximum walking time of 5 minutes to reach the vehicle and 15 minutes to reach the public transit stop/station; for walking and cycling trips, we set a maximum duration of 90 minutes (one and a half hours). After estimating the travel times between CTs, we assumed that all intra-CT trip durations are equal to 0.1 minute.

Commute duration was used to calibrate Probability Density Functions (PDFs) and represent travel impedance, following the approach of Soukhov and Páez (?), Verma and Ukkusuri (?), and Dos Santos et al. [ACTIVE TRAVEL PAPER]. We applied the `fitdistrplus` package (?) and tested different functional forms (uniform, negative exponential, gamma, normal, and lognormal distributions) to select the best PDF for each mode of transportation and municipality, using a selection criteria the lowest Akaike Information Criterion (AIC) value (?) (Appendix 1 displays the estimated impedance functions).

3.3. Modelling employment outcomes

We combined the multimodal spatial availability measures with variables at the individual-level from the NTSP to model perceptions on employment outcomes, assessing two types of weighted regression models. Considering respondent weights, we transformed the continuum perceptions to a likert scale based on their quintile position, categorizing them between: “Most Negative”, “Negative”, “Neutral”, “Positive”, and “Most Positive”. Weighted ordinal logistic regressions were modelled to examine the effect of between the accessibility indicators on the perception categories, using the Equation 3:

$$\log\left(\frac{P(Y \leq j)}{P(Y \geq j)}\right) = \theta_j - \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p \quad (3)$$

Where Y represents the perception category of the respondent (ranging from “Most Negative” to “Most Positive”), and $X_1, X_2, \dots, X_p$ represent the set of explanatory variables, including multimodal spatial availability measures and individual and household characteristic obtained from the NTSP. θ_j and β_p are unknown parameters to be estimated. The parameters β_p are constant for all outcome categories and measure the

change in the log-odds of reporting higher perception categories associated with each explanatory variable, while θ_j represents the intercept separating the perception categories.

Table 2 presents the variables included in the regression models and descriptive statistics (weighted mean and weighted standard deviation for continuous and dummy variables; and weighted count and percentage for categorical variables). We hypothesise that higher values of calculated accessibility will increase the probability of being in more positive perceptions, and that perceptions of accessibility can be connected to the multimodal spatial availability indicators. All other explanatory variables are employed as control variables to properly measure the effect of accessibility indicators on dependent variables.

Table 2: Weighted summary statistics of the explanatory variables.

Explanatory variables	N = 3,684,512
Active transportation job accessibility per user	0.06 (0.13)
Transit job accessibility per user	0.14 (0.09)
Car job accessibility per user	1.00 (0.27)
Active transportation user (dummy)	0.15 (0.36)
Public transit user (dummy)	0.23 (0.42)
Number of Cars	1.30 (0.92)
Number of Bikes	1.15 (1.33)
Preferred mode match most frequent mode	2,794,984 (76%)
Indigenous or racialized person	1,917,721 (52%)
Temporary visa or refugee	272,937 (7.4%)
Person with disability	731,104 (20%)
Woman, transgender or non-binary person	1,828,742 (50%)
Age	40.37 (13.57)
Household size	2.89 (1.61)
Household income	
Less than \$40,000	583,922 (16%)
\$40,000 to \$80,000	1,174,003 (32%)
\$80,000 to \$120,000	965,106 (26%)
More than \$120,000	961,480 (26%)
Population density (people/km2)	7,818.93 (13,582.70)
Population centre class	
Rural	92,135 (2.5%)
Small	48,306 (1.3%)
Medium	59,979 (1.6%)
Large	3,484,092 (95%)
Public transit availability in the neighbourhood	3,537,339 (96%)
Monthly or annual public transit pass	1,083,351 (29%)

¹ Mean (SD); n (%)

3.3.1. Controlling for endogeneity

In social science, it is well understood that “*correlation does not imply causality*” (?). It is important to evaluate the actual impact of a predictor variable (the cause) on the outcome (the effect) when modeling the causal inference. Including causal inference in our modeling approach means we must account for the possible (and indeed likely) presence of endogeneity, which occurs when an independent variable correlates with the regression error term (?) and compromises the consistency of the estimators (?). In this study, we propose to address endogeneity by applying an instrumental variable (IV) technique. An instrumental variable Z must satisfy the following criteria ((??)):

1. Z must be highly correlated with and have a causal effect on the endogenous variable D .
2. Z must affect the dependent variable Y only through D , with no direct effect of Z on Y .
3. Z must not be correlated with the regression residuals of D on Y .

In simpler terms, the instrument must be correlated with the outcome only through accessibility. Examples of instrumental variables used in studies analyzing the relationship between employment outcomes and job accessibility include Euclidean distance from the spatial unit centroid to the nearest major road or employment subcenter ([?]), share of no-vehicle households ([?]), transit accessibility and car unavailability ([?]), distance to rivers ([?; Greg´orio Luz 2022]), and population density ([?; Bastiassen 2026]).

However, two challenges arise in the search for relevant instruments: i) the likelihood that the instruments and error terms are correlated increases as the correlation between the endogenous variable and the instruments becomes stronger, which can render the instruments invalid; and ii) instruments that are only weakly correlated with the endogenous variable may be ineffective and perform poorly. In response to these challenges, Le Gallo and P´az ([?]) developed an approach that relies on synthetic instrumental variables when the model involves spatial data. The researchers demonstrated that synthetic variables can be generated using spatial filtering, a technique that removes spatial residual autocorrelation to improve regression analysis. They propose the use of an eigenvector-based technique, in which transforms a projection of the contiguity matrix into latent map patterns through eigenvector structural analysis. These latent patterns, included in the spatial structure of the system, are then used to generate synthetic variables, which serve as instruments in IV estimation. Since they are derived from the spatial structure of the system, these synthetic instruments are exogenous, thereby fulfilling the core requirement of IV estimation.

Following the methodological approach of Gallo and P´az ([?]) to create spatial filters for the accessibility variables:

1. We built a queen-contiguity matrix for the census tract to represent spatial weights, expressing the existence of a neighbour relation as a binary relationship (1 for neighbours and 0 otherwise).
2. Then, we computed all eigenvectors associated with the contiguity matrix. These eigenvectors are orthogonal to one another and represent a portion of the variance in the spatial contiguity matrix.
3. After this, we initialized a variable S with value 0 and linearly regressed each accessibility indicator (the independent variable) based on each eigenvector and S . If the eigenvector coefficient was significant (adopting a p -value ≤ 0.05), we multiplied the eigenvector by its coefficient and added to value of S . After this process, S represents a spatial filter for the accessibility indicator.

All spatial filters presented a Pearson correlation of above 0.88 with their respective accessibility measures, and, generally, absolute correlations lower than 0.1 with the original perception dependent variables (before transforming them into categorical variables, Appendix A). Since our dependent variables are not linear, the popular technique of replacing endogenous variables by their first-stage predictors (two-stage least square estimation) is not applicable ([?]). Therefore, we applied the two-stage residual inclusion (2SRI) approach. In the first-stage, we estimated the accessibility measures using a linear regression of all control variables plus spatial filters (our instrumental variables). In the second-stage, the dependent variables were estimated using the accessibility measures, the control variables, and the residuals from the first-stage regressions. In both stages, the respondent weights were considered to perform the regressions.

3.4. Equity Standards

We selected the model that best explains the effect of calculated accessibility on the perceptions of accessibility to obtain equity standards. We measured changes in the probabilities across perception categories (from most negative to most positive) by varying the levels of calculated accessibility, while controlling for the other explanatory variables. Controlling for the explanatory variables means holding their values constant, while varying the values of the treatment. Based on this, we decided to evaluate the effects of calculated accessibilities variables in three distinct population profiles:

- An equity-deserving population group, with individual-level control variables set to the values corresponding to populations facing the greatest transportation challenges.
- The most common population, reflecting modal characteristics of the study area population and with control variables set to their most common values; and,
- The advantaged group, defined by selecting for the most privileged population segment in terms of their transport experience.

For each population profile, we increased the value of each accessibility measure by 0.01 jobs per person, while holding constant other accessibility measures (representing alternative transportation modes) at their weighted median values. We also adopted the weighted median value for the population density and a constant zero value for the residuals obtained from the first-stage regressions. Table X presents the control variable values applied for each population group.

Based on the best model and for each population group, we identified the minimum value of calculated job accessibility per person in which the probability of the positive and neutral perceptions combined was higher than the probability of being in negative perceptions, naming this minimum value as a conservative threshold. Next, we defined a progressive threshold as the minimum accessibility value that resulted in the probability of positive perceptions (not considering neutral perceptions) being higher than of negative perceptions. We then identified an intermediate threshold, set as the midpoint between the conservative and progressive thresholds.

Then, we selected as the equity standards the accessibility thresholds obtained based on the equity-deserving population. The rationale behind this equity standard definition is to identify the minimum accessibility level that gives the most disadvantaged population a higher probability of having at least a neutral perception (conservative standard), a neutral to positive perception (intermediate standard), or a positive perception (progressive standard).

3.5. Measuring transportation equity

We calculated the extent and severity of accessibility gap using Foster-Greer-Thorbecke (FGT) measures, according to the Equation 4. FGT measures evaluates the number of people below a standard and the size of the accessibility shortfall relative to it.

$$FGT_{\alpha} = \frac{1}{N} \sum_H^{i=1} \left(\frac{z - y_i}{z} \right)^{\alpha} \quad (4)$$

Where N is the size of the total population; z is the accessibility standard; H is the number of individuals below the z ; y_i is the accessibility level of each individual i ; and α : determines how sensitive the index is to change in the accessibility gap.

Changing the values of α enables the calculation of different FGT indices. When $\alpha = 0$, FGT_0 measures the extent of undersupply as a simple headcount, indicating the proportion of people below the standard. When $\alpha = 1$, FGT_1 measures the severity of the undersupply as the average percent distance between the standard and the accessibility levels of the individuals below it. When $\alpha = 2$, FGT_2 measures both the extent and severity of undersupply by calculating the number of people below the standard, weighted by the size of the accessibility shortfall relative to the standard.

3.6. Areas for priority intervention

For each equity standard, we identified CTs with accessibility shortfall (cases in which the CT accessibility was lower than the equity threshold). Next, we calculated the Accessibility Fairness Index (AFI) (Martens, 2017) to measure the severity of the insufficiency, by multiplying the population size below the standard by the accessibility gap. In practice, the AFI is a simple way to highlight areas with large groups of people that fall the furthest behind the set standards (van der Veen et al., 2020). After this, we classified CTs with accessibility shortfall into levels of priority for policy interventions: Very-high (AFI above the third quartile, top 25%), High (AFI between the median and third quartile, 25-50% from the top), Moderate (AFI between the first quartile and the median, 50-75% from top), and Low priority (AFI below the first quartile, bottom 25%).

4. Results and discussions

4.1. The effect of calculated accessibility on perceived accessibility

Table 3: Models.

Variable	Few jobs		Easy to reach		Travel too long		Decline employment	
	OR	SE	OR	SE	OR	SE	OR	SE
Active transportation job accessibility per user	0.86*	0.064	0.85	0.142	0.95	0.157	0.84*	0.081
Transit job accessibility per user	1.25*	0.112	0.92	0.132	0.93	0.145	0.88	0.184
Car job accessibility per user	0.93	0.097	1.08	0.119	1.11	0.111	1.18	0.139
Number of Cars	0.98	0.121	1.06	0.148	1.04	0.158	1.11	0.136
Number of Bikes	1.08	0.067	0.95	0.093	0.95	0.074	1.02	0.065
Preferred mode match most frequent mode	1.14	0.168	1.72*	0.243	1.68*	0.225	1.30	0.246
Montly or annual public transit pass	0.68*	0.174	1.35	0.220	0.86	0.251	0.52*	0.256
Public transit availability in the neighbourhood	0.75	0.245	0.97	0.361	1.11	0.301	1.06	0.226
Indigenous or racialized person	0.60**	0.164	0.72	0.218	0.71	0.221	0.57**	0.210
Temporary visa or refugee	1.61	0.257	1.00	0.400	0.71	0.359	0.71	0.414
Person with disability	0.69	0.209	0.89	0.275	1.02	0.312	0.80	0.229
Woman, transgender or non-binary person	1.05	0.149	0.84	0.188	1.45	0.203	1.03	0.194
Age	1.01	0.006	1.00	0.008	1.01	0.008	1.02*	0.007
Household size	1.06	0.052	0.96	0.078	1.09	0.087	0.94	0.067
Household income							**	
Less than \$40,000	—	—	—	—	—	—	—	—
\$40,000 to \$80,000	0.80	0.188	1.00	0.257	0.78	0.247	1.48	0.269
\$80,000 to \$120,000	1.23	0.220	1.07	0.257	0.62	0.281	1.96	0.248
More than \$120,000	1.29	0.227	1.37	0.312	0.50	0.305	3.08	0.312
Population density (people/km2)	1.18*	0.075	1.13	0.116	1.28*	0.103	1.10	0.100
Population centre class							***	
Rural	—	—	—	—	—	—	—	—
Small	1.34	0.486	0.68	0.856	1.59	0.815	0.89	0.478
Medium	2.35	0.469	1.41	0.583	3.16	0.571	5.06	0.452
Large	1.24	0.322	1.33	0.438	1.34	0.475	1.16	0.330
Residuals: Active acessibility	1.09	0.078	1.01	0.128	1.03	0.137	0.97	0.085
Residuals: Transit acessibility	0.85	0.092	1.49**	0.130	1.32*	0.118	0.85	0.146
Residuals: Car acessibility	1.13	0.068	0.91	0.127	1.06	0.099	0.98	0.094
Active mode: accessibility $\times$ user	1.17*	0.072	1.10	0.342	1.32	0.359	1.20*	0.088
Transit mode: accessibility $\times$ user	0.95	0.133	1.03	0.169	0.83	0.183	1.17	0.205

Abbreviations: CI = Confidence Interval, OR = Odds Ratio, SE = Standard Error

*p<0.05; **p<0.01; ***p<0.001

4.2. Standards

4.3. Areas for priority intervention

5. Limitations

6. Summary and conclusions

References